

# AI Web Application: Full Stack Architecture & Systems Design

## 1. System Overview

A full-stack enterprise AI chatbot platform supporting multi-turn conversations, Server-Sent Events (SSE) token streaming, user session management, and persistent chat history.

## 2. Technology Stack Breakdown

| Tier           | Technology                           | Key Features                                                |
|----------------|--------------------------------------|-------------------------------------------------------------|
| Frontend       | React 18 + TypeScript + Tailwind CSS | Responsive UI, streaming token renderer, session switcher   |
| Backend API    | FastAPI (Python 3.11) + Uvicorn      | Async request handling, SSE endpoints, Pydantic validation  |
| AI Integration | OpenAI API (GPT-3.5 / GPT-4)         | Token-by-token streaming, custom system prompts             |
| Database       | PostgreSQL + SQLAlchemy ORM          | User authentication, chat session metadata, message history |
| Security       | JWT Auth + SlowAPI Rate Limiter      | Secure access tokens, 10 req/min endpoint rate limiting     |